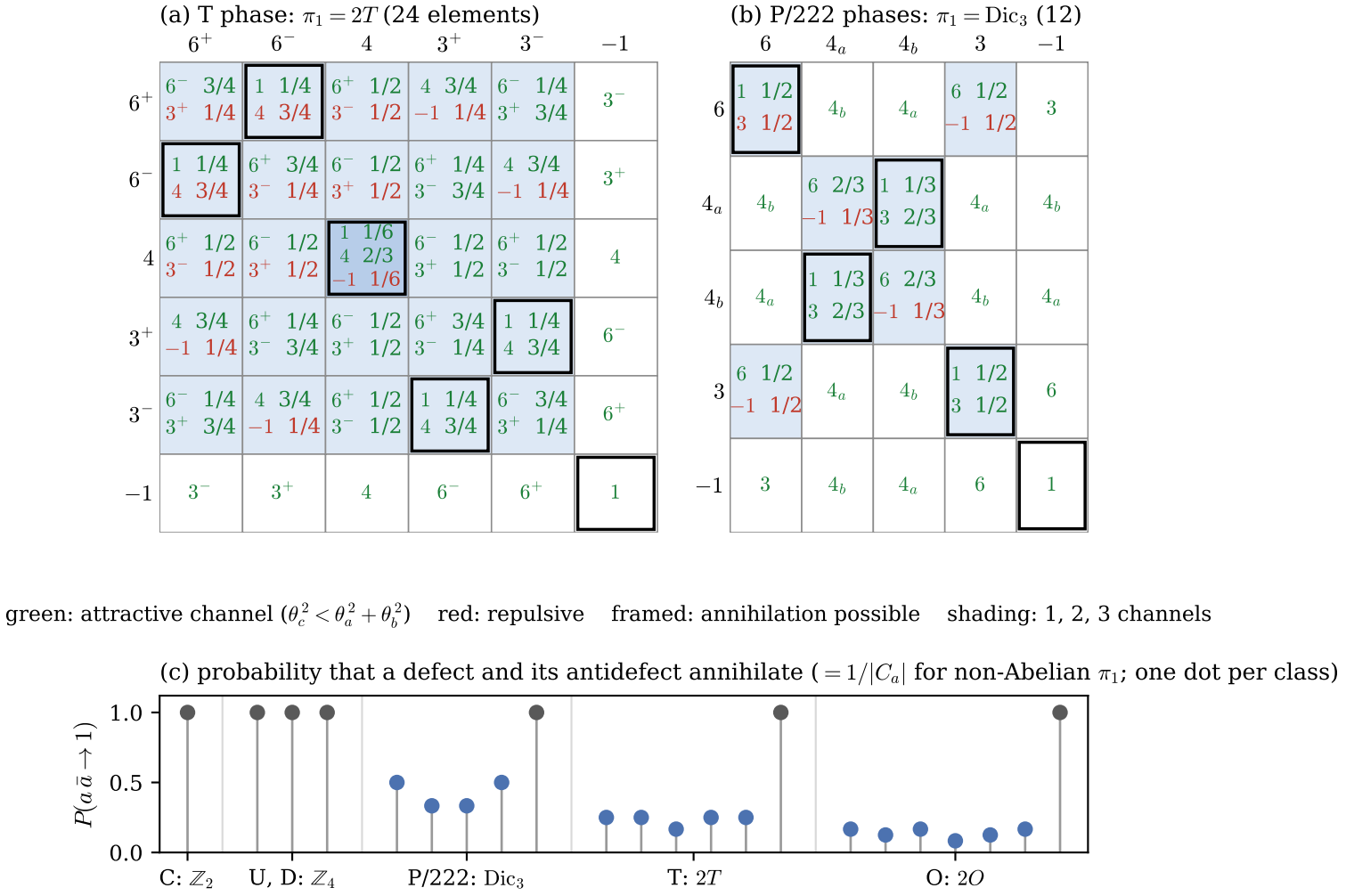
(a) T phase: $\pi_1 = 2T$ (24 elements)
6^+ 6^- 4 3^+ 3^- -1
6^+: 6^- 3/4, 3^+ 1/4 | 1 1/4, 4 3/4 | 6^+ 1/2, 3^- 1/2 | 4 3/4, -1 1/4 | 6^- 1/4, 3^+ 3/4 | 3^-
6^-: 1 1/4, 4 3/4 | 6^+ 3/4, 3^- 1/4 | 6^- 1/2, 3^+ 1/2 | 6^+ 1/4, 3^- 3/4 | 4 3/4, -1 1/4 | 3^+
4: 6^+ 1/2, 3^- 1/2 | 6^- 1/2, 3^+ 1/2 | 1 1/6, 4 2/3, -1 1/6 | 6^- 1/2, 3^+ 1/2 | 6^+ 1/2, 3^- 1/2 | 4
3^+: 4 3/4, -1 1/4 | 6^+ 1/4, 3^- 3/4 | 6^- 1/2, 3^+ 1/2 | 6^+ 3/4, 3^- 1/4 | 1 1/4, 4 3/4 | 6^-
3^-: 6^- 1/4, 3^+ 3/4 | 4 3/4, -1 1/4 | 6^+ 1/2, 3^- 1/2 | 1 1/4, 4 3/4 | 6^- 3/4, 3^+ 1/4 | 6^+
-1: 3^- | 3^+ | 4 | 6^- | 6^+ | 1
(b) P/222 phases: $\pi_1 = \mathrm{Dic}_3$ (12)
6 4_a 4_b 3 -1
6: 1 1/2, 3 1/2 | 4_b | 4_a | 6 1/2, -1 1/2 | 3
4_a: 4_b | 6 2/3, -1 1/3 | 1 1/3, 3 2/3 | 4_a | 4_b
4_b: 4_a | 1 1/3, 3 2/3 | 6 2/3, -1 1/3 | 4_b | 4_a
3: 6 1/2, -1 1/2 | 4_a | 4_b | 1 1/2, 3 1/2 | 6
-1: 3 | 4_b | 4_a | 6 | 1
green: attractive channel ($\theta_c^2 < \theta_a^2 + \theta_b^2$) red: repulsive framed: annihilation possible shading: 1, 2, 3 channels
(c) probability that a defect and its antidefect annihilate ($= 1/|C_a|$ for non-Abelian π_1; one dot per class)
$P(a\,\bar{a} \to 1)$
1.0
0.5
0.0
C: $\mathbb{Z}_2$
U, D: $\mathbb{Z}_4$
P/222: Dic_3
T: $2T$
O: $2O$